Tuo Liu, Ph.D. candidate

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Summary

Personal

I am a Ph.D. candidate in Environmental Health Sciences at the University of Arizona, with a minor in Biostatistics. As a public health researcher with a data scientist mindset, I leverage advanced statistical methods, machine learning, and omics technologies to address complex environmental health challenges. My work focuses on translating high-dimensional data, such as those from metabolomics and proteomics, into actionable insights that enhance our understanding of environmental and occupational exposures, particularly among vulnerable populations like firefighters.

Research

My research focuses on the intersection of omics technologies, statistical analysis, machine learning, and public/occupational health to investigate environmental contaminants and their impacts on human health.
Specifically, I explore occupational exposures, such as those encountered by firefighters, and their links to cancer and other health outcomes. I apply metabolomic and proteomic profiling to examine risk factors for diseases like prostate cancer and long COVID. My work emphasizes the use of cutting-edge statistical and computational approaches to analyze complex, high-dimensional omics data, with the ultimate goal of improving occupational and environmental health outcomes.

Goal

Health Risk Understanding & Mitigation for vulnerable populations, promoting Healthy Aging

Keywords

Metabolomics, Proteomics, Biomarker, Environmental & Occupational Health, Community Engagement, One Health

Education

2021 – Present

Ph.D., University of Arizona
major in **Environmental Health Science**; minor in **Biostatistics**
Thesis title: *Differential Metabolite Expressions in Firefighters Induced by Fireground Exposure: A Comparative Metabolomics Analysis.*

2018 – 2020

M.S., University of Michigan
major in **Environmental Health Science**.
Thesis title: *Real-time personal exposure to PM2.5 and CO among female fish smokers in Ghana: a pilot study.*

2012 – 2016

B.S., China University of Petroleum
major in **Environmental Science**.
Thesis title: *Climate Change Implications of the Ratio of* O^{18}/O^{16} *in the rainwater.*

Research Experience

Firefighter Cancer Cohort Study (FFCCS)

2024 **Female Structure Firefighters** Evaluating Differential Metabolic Profiles by Structure Fire Exposure in Female Firefighters: A Comparison with Male Firefighters.

- Led the collection of 200+ biospecimens for metabolic profiling.
- Conducted differential expression analysis using mass spectrometry on 150 samples.
- Developed and validated statistical models to compare metabolic profiles across firefighter groups.

WUI Fires Evaluating Differential Metabolic Profiles by Wildland-Urban-Interface (WUI) Fire Exposure: A Comparison with Structure Fires.

- Analyzed 85 firefighter samples for WUI-specific metabolic profiles.
- Integrated metabolomics datasets for comparative analysis.
- Presented findings at the 2024 FFCCS Brown Bag meeting.
- Drafted a manuscript (under internal review).

2021-2023 **Male Structure Firefighters** Structure Fires Exposure-Male firefighters: Differential Metabolic Profiles by Hispanic Ethnicity Among Male Tucson Firefighters.

- Developed an analytical pipeline for metabolomics studies.
- Analyzed 100 firefighter samples for ethnicity-specific metabolic profiles.
- Identified biomarkers for differential exposure by Hispanic ethnicity.
- Presented findings at the 2022 Arizona Cancer Center Retreat.
- Authored a manuscript currently under peer review on *Metabolomics*.
- Co-authored a manuscript published on *Scientific Report*.

Arizona Healthcare, Emergency Response, and Other Essential workers Surveillance Study

2024 **Long COVID** Proteomic Profiling to Investigate Biomarkers and Biological Functions Associated with Long COVID in Frontline Workers in the United States (US).

- Led analytical team for proteomics analysis
- Processed and analyzed proteomics data from 130+ frontline and essential workers with long COVID.
- Applied bioinformatics techniques to map 20+ significant biological pathways.
- Leading author for a manuscript currently under development.

2023 **COVID Breakthrough** Serum Proteomics in Omicron SARS-CoV-2 breakthrough: a nested case-control study within a prospective cohort of frontline workers from eight locations in the United States (US).

- Assisted in proteomics analysis for 100+ breakthrough cases to identify risk factors for Omicron breakthrough.
- Contributed to a manuscript currectly under development

Research Experience (continued)

Prostate Cancer risk factors among Prostate Cancer (PCa) Patients

2024 **PCa risk** Evaluating Differential Metabolic Profile by Prostate Cancer Risk among Prostate Cancer Patients.

- Collaborated with Chemical Analytics team and collected metabolic data from 22 prostate cancer patients.
- Identified metabolites and metabolic functions significantly associated with high-risk prostate cancer.
- Leading draft of a manuscript.

Particulate Matter (PM) Exposure among Female Fish Smokers in Ghana

2020 **PM** Evaluating real-time exposure to PM and carbon monoxide of female fish smokers; assess exposure-reducing efficiency of newly developed stove as compared to the traditional stove; evaluate the validity of using CO as a exposure proxy for PM.

- Led fieldwork for real-time monitoring of PM and CO in 120+ households in Ghana.
- Evaluated the effectiveness of newly developed stoves regarding reduction of PM exposure.
- Demonstrated the validity of using CO as a proxy for PM in over 100 exposure assessments.
- Contributed to a report and drafted a thesis that was reported back to participants.

Publications

Journal Articles

1. M. A. Furlong, **T. Liu**, A. Jung, *et al.*, "Per- and polyfluoroalkyl substances (pfas) and microrna: An epigenome-wide association study," *manuscript under internal review*, 2024.
2. **T. Liu**, k. Batai, J. M. Snider, M. A. Furlong, and J. L. Burgess, "Evaluating differential metabolic profile by prostate cancer risk among prostate cancer patients," *manuscript under preparation*, 2024.
3. **T. Liu**, M. A. Furlong, J. M. Snider, D. I. Walker, and J. L. Burgess, "Evaluating differential metabolic profile by wildland-urban-interface fire exposure: A comparison with structure fire," *manuscript under internal review*, 2024.
4. **T. Liu**, M. A. Furlong, J. M. Snider, *et al.*, "Differential metabolic profiles by hispanic ethnicity among male tucson firefighters," *Metabolomics, under peer review*, 2024.
5. **T. Liu**, D. Stea, M. A. Furlong, and J. L. Burgess, "Proteomic profiling to investigate biomarkers and biological functions associated with long covid in frontline workers," *manuscript under preparation*, 2024.
6. M. A. Furlong, **T. Liu**, J. M. Snider, *et al.*, "Evaluating changes in firefighter urinary metabolomes after structural fires: An untargeted, high resolution approach," *Scientific Reports*, vol. 13, no. 1, p. 20 872, 2023, ISSN: 2045-2322. DOI: 10.1038/s41598-023-47799-x.

Miscellaneous Experience

Awards and Achievements

2023 **One Health GRA Award: $50,000 (MEZCOPH)**, University of Arizona.

Miscellaneous Experience (continued)

2021 **Dean Named Scholarship, $10,000**, University of Arizona.

Mentorship and service

May-Aug, 2024 **Graduate-Mentor for the Undergraduate Research Opportunities Consortium (UROC)-Summer Research Institute (SRI)**, University of Arizona.

2022-2024 **Volunteer for Graduate Orientation**, University of Arizona.

Talks

July, 2024 **Differential Metabolic Profiles by WUI Fires: A Comparison With Structure Fires.**, FFCCS Data Brown Bag Meeting.

April, 2022 **Differential Metabolites by Structure Fire Exposure in Male Firefighters.**, Arizona Cancer Center Scientific Retreat.

Certification

2024 **Deep Learning Specification**. Awarded by Coursera.

2020 **IBM Data Science Professional Certificate**. Awarded by Coursera.

Statistics & Data Science Projects

Statistical Techniques

2022 **Applied Biostatistics**

- Longitudinal analysis
- Survey analysis
- Propensity matching analysis
- Survival analysis

Data Science Projects

2023 **Arizona Pesticide Usage Map**.

- Built a Shiny application with R to explore pesticide use in the State of Arizona, 1992-2016.
- Visualized pesticide usage with an interactive map & chart using Leaflet & Plotly.

Expense Tracker

- Built a web application using Streamlit.
- Connected to and stored customer data in NoSQL database from deta.
- Visualized personal expense with an interactive Sankey chart using Plotly.

2020 **Exploring neighborhoods & Chinese Restaurants in Toronto**.

- Defined a business problem, searched for open source data to answer the question.
- Applied machine learning algorithms to identify biomarkers related to Hispanic ethnicity.
- Used Foursquare API to compare different neighborhoods of Toronto to evaluate the distribution of Asian & Chinese restaurants.
- Determined which neighborhood is most suitable for starting a new Chinese restaurant.

Skills

Languages	Strong reading, writing and speaking, work fluent in English, native in Mandarin Chinese.
Coding	Python, R, SQL, LaTeX, …
Databases	MySQL, PostgreSQL.
Misc.	Academic research, training, consultation, LaTeX typesetting and publishing.

References

Available on Request